Disk management

Disk management in Linux refers to the process of monitoring, organizing, and controlling storage devices such as hard disks, SSDs, and their partitions.

Hands-On: Disk Management Commands in Linux

Image 1: Listing Block Devices

lsblk

- Displays all available block devices such as disks and partitions
- Shows device name, size, type, and mount points

```
        _/m/'
[ec2-user@ip-172-31-39-116 ~]$ lsblk
NAME           MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1        259:0    0   8G  0 disk
├─nvme0n1p1    259:1    0   8G  0 part /
├─nvme0n1p127  259:2    0   1M  0 part
└─nvme0n1p128  259:3    0  10M  0 part /boot/efi
[ec2-user@ip-172-31-39-116 ~]$
```

Image 2: Checking Disk Usage

df -h

- Displays disk space usage of mounted file systems
- Shows total size, used space, available space, and mount points in a human-readable format

```
[ec2-user@ip-172-31-39-116 ~]$ df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             459M     0  459M   0% /dev/shm
tmpfs             184M  432K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G  6.4G  20% /
tmpfs             459M     0  459M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
[ec2-user@ip-172-31-39-116 ~]$
```

Image 3: Checking Memory Usage

free -h

- Displays system memory and swap usage
- Shows total, used, free, and available memory in a human-readable format

```
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$ free -h
              total        used        free      shared  buff/cache   available
Mem:           916Mi       167Mi       525Mi       0.0Ki       223Mi       618Mi
Swap:             0B          0B          0B
```

Image 4: Viewing Network Interface Details

ifconfig

- Displays network interface information such as IP address, MAC address, and status
- Used to check network configuration and connectivity

```
[ec2-user@ip-172-31-39-116 ~]$ ifconfig
ens5: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 9001
        inet 172.31.39.116  netmask 255.255.240.0  broadcast 172.31.47.255
        inet6 fe80::11:24ff:fe93:c8c7  prefixlen 64  scopeid 0x20<link>
        ether 02:11:24:93:c8:c7  txqueuelen 1000  (Ethernet)
        RX packets 8689  bytes 57789073 (55.1 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1934  bytes 188173 (183.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 12  bytes 1020 (1020.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 12  bytes 1020 (1020.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Image 5: Checking Listening Ports

ss -tuln

- Displays all **listening TCP and UDP ports**
- Shows port numbers without resolving service names
- Useful for checking active services and open ports on the system

```
ec2-user@ip-172-31-39-116 ~]$ ss -tuln
etid          State          Recv-Q          Send-Q                                  Local Address:Port                  Peer Address:Port

dp            UNCONN         0               0                                   172.31.39.116%ens5:68                       0.0.0.0:*

dp            UNCONN         0               0                                           127.0.0.1:323                       0.0.0.0:*

dp            UNCONN         0               0                        [fe80::11:24ff:fe93:c8c7]%ens5:546                        [::]:*

dp            UNCONN         0               0                                             [::1]:323                         [::]:*

cp            LISTEN         0               128                                         0.0.0.0:22                          0.0.0.0:*

cp            LISTEN         0               128                                            [::]:22                          [::]:*

ec2-user@ip-172-31-39-116 ~]$
```

Image 6: Testing Network Connectivity

ping google.com

- Tests network connectivity between the system and a remote host
- Displays response time and packet loss information

```
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$ ping google.com
PING google.com (142.250.77.46) 56(84) bytes of data.
64 bytes from bom07s26-in-f14.1e100.net (142.250.77.46): icmp_seq=1 ttl=117 time=1.58 ms
64 bytes from bom07s26-in-f14.1e100.net (142.250.77.46): icmp_seq=2 ttl=117 time=1.59 ms
64 bytes from bom07s26-in-f14.1e100.net (142.250.77.46): icmp_seq=3 ttl=117 time=1.60 ms
64 bytes from bom07s26-in-f14.1e100.net (142.250.77.46): icmp_seq=4 ttl=117 time=1.59 ms
64 bytes from bom07s26-in-f14.1e100.net (142.250.77.46): icmp_seq=5 ttl=117 time=1.58 ms
64 bytes from bom07s26-in-f14.1e100.net (142.250.77.46): icmp_seq=6 ttl=117 time=1.59 ms
64 bytes from bom07s26-in-f14.1e100.net (142.250.77.46): icmp_seq=7 ttl=117 time=1.58 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6012ms
rtt min/avg/max/mdev = 1.577/1.588/1.602/0.007 ms
[ec2-user@ip-172-31-39-116 ~]$
```

Image 7: Downloading Files from the Internet

wget https://example.com/file

- Downloads files from the internet using the terminal
- Commonly used to fetch packages, scripts, or resources on Linux systems

```
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$ wget https://cdn.pixabay.com/photo/2022/09/27/19/46/ai-generated-7483596_960_720.jpg
--2026-03-04 02:19:38--  https://cdn.pixabay.com/photo/2022/09/27/19/46/ai-generated-7483596_960_720.jpg
Resolving cdn.pixabay.com (cdn.pixabay.com)... 104.18.40.96, 172.64.147.160, 2606:4700:4400::ac40:93a0, ...
Connecting to cdn.pixabay.com (cdn.pixabay.com)|104.18.40.96|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 256853 (251K) [image/jpeg]
Saving to: 'ai-generated-7483596_960_720.jpg'

ai-generated-7483596_960_720.jpg       100%[===================================================================>] 250.83K  --.-KB/s    in 0.003s


2026-03-04 02:19:38 (77.6 MB/s) - 'ai-generated-7483596_960_720.jpg' saved [256853/256853]

[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$ ls
ai-generated-7483596_960_720.jpg
[ec2-user@ip-172-31-39-116 ~]$
```

Image 8: Viewing Running Processes

ps -eof

- Displays all running processes in detailed format
- Shows process ID (PID), user, CPU usage, memory usage, and command
- Useful for monitoring and managing system processes

```
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$ ps -eof
F
4
1
1
1
1
```

Image 9: Viewing All Running Processes

ps -ef

- Displays all running processes in full format
- Shows process owner, PID, parent PID, and command
- Useful for process monitoring and troubleshooting

```
[ec2-user@ip-172-31-39-116 ~]$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 02:08 ?        00:00:00 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root           2       0  0 02:08 ?        00:00:00 [kthreadd]
root           3       2  0 02:08 ?        00:00:00 [rcu_gp]
root           4       2  0 02:08 ?        00:00:00 [rcu_par_gp]
root           5       2  0 02:08 ?        00:00:00 [slub_flushwq]
root           6       2  0 02:08 ?        00:00:00 [netns]
root           8       2  0 02:08 ?        00:00:00 [kworker/0:0H-events_highpri]
root          10       2  0 02:08 ?        00:00:00 [mm_percpu_wq]
root          11       2  0 02:08 ?        00:00:00 [rcu_tasks_kthread]
root          12       2  0 02:08 ?        00:00:00 [rcu_tasks_rude_kthread]
root          13       2  0 02:08 ?        00:00:00 [rcu_tasks_trace_kthread]
root          14       2  0 02:08 ?        00:00:00 [ksoftirqd/0]
root          15       2  0 02:08 ?        00:00:00 [rcu_preempt]
root          16       2  0 02:08 ?        00:00:00 [migration/0]
root          18       2  0 02:08 ?        00:00:00 [cpuhp/0]
root          19       2  0 02:08 ?        00:00:00 [cpuhp/1]
root          20       2  0 02:08 ?        00:00:00 [migration/1]
root          21       2  0 02:08 ?        00:00:00 [ksoftirqd/1]
root          23       2  0 02:08 ?        00:00:00 [kworker/1:0H-events_highpri]
root          26       2  0 02:08 ?        00:00:00 [kdevtmpfs]
root          27       2  0 02:08 ?        00:00:00 [inet_frag_wq]
root          28       2  0 02:08 ?        00:00:00 [kauditd]
root          29       2  0 02:08 ?        00:00:00 [khungtaskd]
root          30       2  0 02:08 ?        00:00:00 [oom_reaper]
```

Image 10: Top Memory-Consuming Processes

ps -eo %cpu,%mem --sort=%mem | head -n 10

- Displays processes sorted by **memory usage**
- Shows top processes consuming the most memory
- Useful for performance monitoring and system optimization

```
[ec2-user@ip-172-31-39-116 ~]$ ps -eo %cpu,%mem --sort=-%mem | head -n 10
%CPU %MEM
 0.0  2.0
 0.0  1.8
 0.0  1.6
 0.0  1.6
 0.0  1.4
 0.0  1.2
 0.0  1.0
 0.0  1.0
 0.0  1.0
[ec2-user@ip-172-31-39-116 ~]$
```

Image 11: Real-Time System Monitoring

top

- Displays real-time information about system processes
- Shows CPU usage, memory usage, and running processes
- Useful for monitoring system performance and resource utilization

```
top - 02:50:36 up 42 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 100 total,   1 running,  99 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    916.8 total,    461.2 free,    166.9 used,    288.8 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    618.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  172408  17404  10852 S   0.0   1.9   0:00.92 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.06 rcu_preempt
     16 root      rt   0       0      0      0 S   0.0   0.0   0:00.01 migration/0
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
```

Image 12: Advanced System Monitoring Using htop

sudo yum install htop

htop

- Installs **htop**, an interactive process monitoring tool
- Displays CPU, memory usage, and running processes in a user-friendly interface
- Allows easy process management using keyboard shortcuts

```
[root@ip-172-31-39-116 ec2-user]# yum install htop
Last metadata expiration check: 0:30:16 ago on Wed Mar  4 02:21:39 2026.
Dependencies resolved.
================================================================================================================================
 Package                    Architecture                Version                                  Repository                  Size
================================================================================================================================
Installing:
 htop                       x86_64                      3.2.1-87.amzn2023.0.3                    amazonlinux                183 k

Transaction Summary
================================================================================================================================
Install  1 Package

Total download size: 183 k
Installed size: 432 k
Is this ok [y/N]: y
Downloading Packages:
htop-3.2.1-87.amzn2023.0.3.x86_64.rpm                                                                 6.0 MB/s | 183 kB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                 2.5 MB/s | 183 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                          1/1
  Installing       : htop-3.2.1-87.amzn2023.0.3.x86_64                                                                        1/1
  Running scriptlet: htop-3.2.1-87.amzn2023.0.3.x86_64                                                                        1/1
  Verifying        : htop-3.2.1-87.amzn2023.0.3.x86_64                                                                        1/1

Installed:
  htop-3.2.1-87.amzn2023.0.3.x86_64

Complete!
```

```
  0[||||||||||||||||||||||||||||||                          42.2%] Tasks: 38, 15 thr, 72 kthr; 2 running
  1[||||||||||||||||||||||||||||||                          42.2%] Load average: 0.30 0.07 0.02
Mem[||||||||||||||||||||||||||||||||||||||||||||||||||||| 189M/917M] Uptime: 00:44:13
Swp[                                                        0K/0K]

  Main   I/O
    PID USER      PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
  3267 root       39  19  220M  6020  2360 R  33.3  0.6  0:00.44 /usr/bin/mandb -q
      1 root       20   0  168M 17404 10852 S   0.0  1.9  0:00.94 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
    827 root       20   0  53548 15020 13876 S   0.0  1.6  0:00.22 /usr/lib/systemd/systemd-journald
   1240 root       20   0  31996 11568  8520 S   0.0  1.2  0:00.06 /usr/lib/systemd/systemd-udevd
   1311 systemd-re 20   0  22572 15120 11628 S   0.0  1.6  0:00.06 /usr/lib/systemd/systemd-resolved
   1313 root       16  -4  21192  2252  1500 S   0.0  0.2  0:00.01 /sbin/auditd
   1314 root       16  -4  21192  2252  1500 S   0.0  0.2  0:00.00 /sbin/auditd
   1403 root       20   0  16372  6456  5572 S   0.0  0.7  0:00.00 /usr/bin/systemd-inhibit --what=handle-suspend-key:handle-hibernate-key --who=noah --w
   1406 root       20   0  81428  1572  1440 S   0.0  0.2  0:00.08 /usr/sbin/irqbalance --foreground
   1407 libstorage 20   0   2780  1992  1828 S   0.0  0.2  0:00.01 /usr/bin/lsmd -d
   1410 root       20   0  16876  7952  6932 S   0.0  0.8  0:00.02 /usr/lib/systemd/systemd-homed
   1411 root       20   0  18712 10064  7836 S   0.0  1.1  0:00.06 /usr/lib/systemd/systemd-logind
   1415 root       20   0  81428  1572  1440 S   0.0  0.2  0:00.00 /usr/sbin/irqbalance --foreground
   1433 dbus       20   0   8492  4136  3532 S   0.0  0.4  0:00.02 /usr/bin/dbus-broker-launch --scope system --audit
F1Help  F2Setup F3Search F4Filter F5Tree  F6SortBy F7Nice -F8Nice +F9Kill  F10Quit
```

Image 13: Tracing Network Route to a Public IP

traceroute 8.8.8.8

- Displays the path (hops) taken by packets to reach a destination
- Shows intermediate routers and response times
- Useful for troubleshooting network delays and connectivity issues

```
[ec2-user@ip-172-31-39-116 ~]$
[ec2-user@ip-172-31-39-116 ~]$ traceroute 15.206.28.212
traceroute to 15.206.28.212 (15.206.28.212), 30 hops max, 60 byte packets
 1  244.5.0.153 (244.5.0.153)  7.901 ms * 244.5.0.173 (244.5.0.173)  4.934 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * *
```